

Leveling Length-Based Control Guide

x-axis = coil length (m) · ERP/PLC joined by coil ID · tension joined by time

09/03/2026

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EPD2871C01 · Daehan Steel · 0.76X4XC

▼

Same-spec history: 3 coils

● RUNNING

Purpose and how the data is joined

The goal of this page is to answer one question for the operator and the manager: **"At this point in the coil, how should the line be controlled?"** — tension output, line speed, and when to step them down as the coil gets smaller. To do that, three sources are combined:

- **ERP work order** (erp_data, MariaDB) — joined by **coil ID**. Gives customer, spec (thickness × width), weight, status. Weight + spec are converted into the coil's total length, which becomes the x-axis.
- **PLC log** (LEVELING_DATA, MariaDB, one row every ~2 s) — joined by **coil ID**. Gives cut length setting, per-stroke movement, inverter load, cycle time. Stroke resets are counted to compute the cumulative length processed so far.
- **Tension controller readings** (ai_tension_readings, Supabase, camera + AI every 2–5 min) — joined by **time**: a reading is attributed to the coil that was running at captured_at, and placed on the length axis at the cumulative length reached at that moment.

Everything below uses today's real values (2026-09-03, KST). The purple boxes are annotations for review, not part of the final page. The "same-spec history" recommendations are **illustrative** — see the honesty note in the guide section.

Coil erp_data · by COIL_ID

EPD2871C01

Daehan Steel Co. · 0.76 × 4 ft (1,219 mm) × Coil · 10,410 kg

Total length (derived) ≈ 1,432 m (weight ÷ thickness × width × 7.85)

Cut length 2,994 mm → ≈ 478 sheets expected

Start diameter (controller) 1,204 mm

Run started 08:33 (PLC) · 08:25 (ERP)

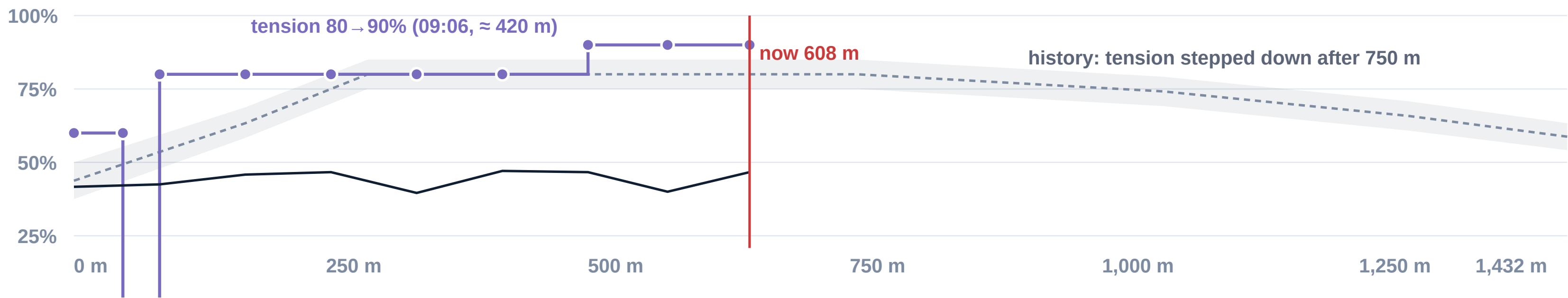
Total length. Leveling has no coil-length sensor, so the full length is derived from ERP: 10,410 kg ÷ (0.076 cm × 121.9 cm × 7.85 g/cm³ = 7.27 kg/m) ≈ 1,432 m. The width token "4자" (4 ft) in the spec string is parsed to 1,219 mm. Expected sheets = total length ÷ cut length.

Current position cumulative length = real strokes × cut length · stroke = MOVELENSV reset

Length processed 608 _m 203 sheets × 2.994 m · 42% done		Remaining 824 _m ≈ 275 sheets · ETA 11:10	Remaining coil Ø (est.) ≈ 900 _{mm} start 1,204 → core 508	Real time per sheet 19.5 _s stroke 9.4 s + feed/stop 10 s	
0–250 m · Start		250–750 m · Steady	750–1,200 m · Late		1,200 m+ · Tail
tension 60 → 80%		tension 90% · 19.5 s/sheet	history: tension ↓ 85%		history: 75% · slow

How position is computed — and a correction to the existing screens. The PLC has no cumulative-length column. `MOVELENSV` climbs from 0 toward the cut length on every stroke and drops back to 0 when the sheet is cut; each drop is counted as one sheet, × `CUTLENSET` = length processed. The stroke count shown by the current tools (966) simply counts 2-second rows where `CYCLETM` > 0; since `CYCLETM` is a counter that rises 0 → 9.4 s within one stroke, each sheet is counted 3–4 times. Today's log shows a real sheet takes ≈ 19.5 s (9.4 s stroke + ≈ 10 s feed/stop), i.e. ≈ 203 sheets / 608 m by 09:40 — consistent with the derived 1,432 m total. This page uses the reset-based count. Remaining diameter is estimated from remaining length, thickness and the 508 mm core.

Length profile x = cumulative length (m) · tension placed by time → length mapping · grey = same-spec history



- This coil — tension output % (readings, time → length) ■ Same-spec history — mean tension (n=3, band = min-max)
- This coil — seconds per sheet (secondary scale 10–30 s) ■ Current position

Join rule in detail. ERP and PLC rows share `COIL_ID`. Each tension reading's `captured_at` is matched to the coil with `RUN=1` at that time, and to the number of sheets cut up to that moment × cut length = its position on the x-axis. Readings arrive every 2–5 minutes, so on the length axis they are 40–120 m apart; between readings the line is drawn as a step (last setting holds). The history band is built from completed coils with the same thickness (±0.03 mm) and width, each normalised to 0–100 % of its own length before averaging, so coils of different weights can be compared.

Control guide by length segment recommended (same-spec history) vs. current setting

0 – 250 m Start (large Ø)	Rec. tension 70–80%	Rec. speed 25 m/min	Applied 60 → 80%
Done · matches history. The initial 60% is the set-up window right after the coil change.			
250 – 750 m Steady · current	Rec. tension 85–90%	Rec. speed 25 m/min	Current 90% OK
19.5 s/sheet equals the history mean (19.8 s). Hold.			
750 – 1,200 m Late (Ø shrinking)	Rec. tension 85% → 80%	Rec. speed 25 m/min	Current 90% Prepare
All 3 history coils stepped down near 750 m. As the remaining diameter shrinks, the same output % means higher real strip tension → risk of camber / scratches.			
1,200 – 1,432 m Tail	Rec. tension 75%	Rec. speed 20 m/min	Current —
Last ~100 m ran slow and low-tension in history. 25 s+/sheet is normal while handling the tail.			

Where the recommendations come from — and their current limit. Phase 1 shows, per length segment, the **median tension and speed actually used on previous coils of the same spec** (same approach as the slitting AI Helper's `recommend_for` mode). If this coil's cycle time or inverter load is worse than history, the guide suggests one step more conservative. **Honest caveat:** tension readings only exist since 2 Sep and only 11 leveling coils have ever been tagged with a coil ID, so there is not yet enough history to produce per-spec recommendations. Until coils accumulate, the guide shows this coil's own measured profile plus "setting changed here → cycle time afterwards", and the history band/recommendations fill in automatically as data grows. The values in this mockup's guide are examples of the final form, not measured recommendations.

Tension reading → length mapping ai_tension_readings · by time

Time	Length	Out %	μm	Segment
08:31	—	60	800	before run (prev. coil)
08:36	0 m	0	760	coil change
08:41	≈ 36 m	80	760	Start
08:51	≈ 128 m	80	760	Start
09:01	≈ 258 m	80	760	Steady
09:06	≈ 420 m	90	760	Steady changed
09:11	≈ 466 m	90	760	Steady
09:21	≈ 608 m	90	760	Steady · now

Audit table. Every reading with the length it was attributed to, so the time-based join can be checked by eye. Readings before the first RUN=1 row are labelled "before run"; 0 % output is labelled "coil change".

Optional steps that make the mapping exact:

- ① Have ocr_monitor_production.py store the running COIL_ID and the sheet count at capture time — removes the need for time matching.
- ② If the controller's tension_rate (currently always read as 0.0) or remaining diameter becomes readable, length can be cross-checked without the PLC.
- ③ Collecting a cut-counter tag from the PLC turns the reset-based estimate into a measured value.

Implementation plan (after approval)

- File:** public/leveling-length-guide.html (vanilla JS, inline styles, Pretendard via jsDelivr, same conventions as the other public/*.html tools) + a card in LabPage.jsx under 레벨링 공정 관리 with external: true. Office-PC layout; an EN mirror can follow the KR/EN pattern used by the slitting tools.
- Backend:** one new mode in leveler-explore, e.g. leveling_length_profile=<coil>&leveling_work_date=<date>, that returns the reset-based sheet count over time (timestamp of each MOVELENSV drop) plus per-sheet cycle and load. A second mode, leveling_history=<thickness>&width=..., returns length-normalised profiles of completed same-spec coils. Everything else reuses existing modes and the Supabase anon read of ai_tension_readings.
- Client logic:** total length from ERP weight/spec; time → length mapping for tension readings; segment boundaries at 0 / 250 / 750 / 1,200 m by default (configurable, or as % of total length); recommendation = per-segment median of history, with a conservative step if current cycle/load is worse.
- Refresh:** 20 s. KST "today" via the double-offset-safe pattern; MariaDB timestamps treated as KST wall-clock.
- Out of scope for phase 1:** changing the OCR script (phase 2, improves accuracy), mobile layout, login.

Sources: erp_data / LEVELING_DATA (MariaDB via leveler-explore) · ai_tension_readings (Supabase) · times in KST · mockup — real 2026-09-03 values; history recommendations illustrative